

ALLEN STIVANSON CHRISTIAN

AI & Data Science Engineer · Patent Holder · Python Backend Developer

+91 6355941155 | cstivanson@gmail.com | allenschristian07@gmail.com | Anand, Gujarat, India
github.com/allen745 | linkedin.com/in/allen-christian-708545409 | instagram: @msd7_allen1405

PROFESSIONAL SUMMARY

Motivated AI & Data Science engineering student at AD Patel Institute of Technology with a Government of India registered design patent (Design No. 450815-001, issued July 2025). Experienced in Python, FastAPI, SQL, machine learning, IoT, and robotics. Co-author of a published research paper on piezoelectric energy harvesting. Competed at national-level tech events including TCS Rural IT Quiz and AIKYAM 1.0 National Techfest. Completed an AI internship at MY JOB GROW. Eager to contribute to data-driven, intelligent systems.

EDUCATION

B.Tech — Artificial Intelligence & Data Science | [AD Patel Institute of Technology \(ADIT\), CVM University](#)
2025 – 2029 (Expected)

New Vallabh Vidyanagar, Anand, Gujarat

High School Diploma — Science Stream | [Charotar English Medium School \(CEMS\)](#) Completed April 2025

Managed by Charotar Education Society, Anand, Gujarat · Principal: Shri Kartikbhai Y. Patel

- Multiple national-level competition participations and a government-registered patent during school

WORK EXPERIENCE

AI Internship Student | [MY JOB GROW](#)

November 2025 – February 2026

Gujarat, India

- Completed structured training in AI processes, tools, and methodologies
- Participated in on-the-job training applying AI theory to real-world project tasks
- Contributed to live AI-based project solutions with hands-on technical implementation
- Gained practical experience with Python-based AI tools, data pipelines, and automation

TECHNICAL SKILLS

Languages: Python (Advanced), SQL, JavaScript (basic)

Backend: FastAPI, REST API design, Pydantic, Uvicorn, SQLAlchemy, JWT Authentication, Async/Await

Databases: MySQL, SQLite, Query Optimization, Joins & Aggregations, Relational Schema Design

AI / ML: Scikit-learn, NumPy, Pandas, OpenCV, MediaPipe, Computer Vision, Model Evaluation & Pipelines

Visualization: Matplotlib, Seaborn, Exploratory Data Analysis (EDA), Statistical Profiling

Hardware / IoT: Arduino, ESP32-S3, OpenMV H7, Mecanum 4WD, Piezoelectric Systems, Sensor Integration

Tools & Platforms: Git, GitHub, VS Code, Jupyter Notebook, Google Colab, Linux basics

Other: Market & Finance Analysis, A* Pathfinding Algorithm, Cybersecurity (CTF basics)

PROJECTS

Spring-Loaded Piezoelectric Generating Striker Device | [Patented](#) · [GOI Design No. 450815-001](#)

Filed Feb 2025 · Issued Jul 2025

- Co-inventor: spring-loaded mechanism using piezoelectric transducers to generate electrical energy on mechanical impact
- Registered in Class 13-01 at The Patent Office, Government of India · Serial No. 205545
- Filed 27 February 2025 · Registered 08 March 2025 · Issued 28 July 2025
- Co-authored research paper: design, methodology, and applications of the device
- Team: Patel Yug, Kariya Hitansh, Maji Dev, Pathaan Shahnawazkhan, Christian Allen Stevenson · Mentors: Shah Neel Atulkumar, Gandhi Keval Vipulkumar

TrackBot — Warehouse Logistics Robot | [Robotics](#) · [AI Navigation](#) · [IoT](#)

2025 – 2026

- Autonomous guided vehicle (AGV) with A* pathfinding algorithm for dynamic obstacle avoidance
- Hardware: ESP32-S3 microcontroller + OpenMV H7 for onboard computer vision, Mecanum 4WD for omnidirectional movement
- SSIP (Student Startup & Innovation Policy) funding proposal submitted to Gujarat government

Hand Gesture Recognition Model | [Machine Learning](#) · [Computer Vision](#) · [Python](#)

2025

- Real-time hand gesture detection using MediaPipe landmark extraction + custom trained classifier
- Enables touchless human-computer interaction; applicable to accessibility tools and HCI interfaces

AutoSeed Bot | [IoT](#) · [Robotics](#) · [Agricultural Technology](#)

2024

- Autonomous agricultural robot with IR/ultrasonic sensor navigation and precision seed dispersion mechanism
- Addresses labour inefficiency in small-scale farming through IoT-driven smart automation

ACHIEVEMENTS & COMPETITIONS

-
- ▶ **2nd Position — Model Presentation, Skill Fair (Chatkaro-2024)** 13 January 2024
Charotar Education Society Chatkaro-2024 · Held 11–13 January 2024 · Signed: Shri Niravkumar N. Patel (Chairman) & Shri Ketankumar P. Patel (Secretary)
 - ▶ **Certificate of Appreciation — Patent Achievement** 09 January 2026
Awarded for successful registration of Spring-Loaded Piezoelectric Generating Striker Device · Signed: Principal Kartikumar Patel, CEMS
 - ▶ **Participation — Science Exhibition (Chatkaro-2025)** 13 January 2025
"Endless Awakening Through Discovery" Science Exhibition · Held 11–13 January 2025 · Signed: Shri Niravkumar N. Patel & Shri Ketankumar P. Patel
 - ▶ **CVM University Hackathon 2026 — Participant** 6–7 March 2026
Under Institution's Innovation Council (IIC) & SSIP · Signed: Dr. Dinesh Prajapati, Mr. Chetan Verma (Coordinators), Dr. Yashavant Patel (Co-Coordinator, CVMUISC), Prof. (Dr.) Vishal Singh (Convener / Principal ADIT)
 - ▶ **SPEC Innovation Award — AIKYAM 1.0 National Techfest** 7–8 February 2024
Project Exhibition under SPEC Innovation Award · Managed by Tirupati Foundation Trust · Signed: Dr. Rutvij Joshi (Principal) & Prof. Jigisha Trivedi (Co-ordinator)
 - ▶ **TCS Rural IT Quiz 2024 — National Level Participant** 2024
Organised by TATA Consultancy Services & Dept. of Electronics, IT, Bt, Science & Technology, Govt. of Karnataka · Signed: Darshan H.V. IAS (Director, Karnataka Innovation & Technology Society) & Sunil Deshpande (Regional Head, TCS Bengaluru)
 - ▶ **Rural IT Quiz 2024 — District Level** 4 September 2024
GUJCOST-sponsored, Vallabh Vidyanagar, Dist. Anand · Signed: Dr. Vibha Vaishnav, Hon. Director, CCP CSC
 - ▶ **Rural IT Quiz 2023 — District Level** 22 August 2023
Supported by GUJCOST, Gandhinagar under DST, GOG · Vallabh Vidyanagar, Dist. Anand · Signed: Dr. Vibha Vaishnav, Hon. Director, CCP CSC
 - ▶ **HackZero'26 — CTF Competition (Team: princepatel131984)** 28–29 March 2026
2-day Capture The Flag event organised by OWASP VIT Bhopal University · Tackled web exploitation, cryptography, forensics · Signed: Dr. D. Saravanan, Faculty Advisor

CERTIFICATIONS

- Gemini for Google Workspace — *Simplilearn SkillUP · Powered by Google Cloud · Code: 1024848420* May 2026
- AI Fundamentals & Tools (Internship) — *MY JOB GROW, Gujarat* Nov 2025 – Feb 2026
- Python for Data Science — *Online Certification* 2025
- Machine Learning Foundations — *Online Certification* 2025
- SQL for Data Analysis — *Online Certification* 2025
- FastAPI — Building APIs with Python — *Online Certification* 2026
- Data Visualization with Python — *Online Certification* 2025
- Introduction to Deep Learning — *Online Certification* 2025
- Computer Vision with OpenCV — *Online Certification* 2025
- IoT & Embedded Systems Basics — *Online Certification* 2024
- Git & GitHub for Developers — *Online Certification* 2024

PATENT & RESEARCH

Spring-Loaded Piezoelectric Generating Striker Device

Design No. 450815-001 · Serial No. 205545 · Class 13-01 · The Patent Office, Government of India

Filed: 27 February 2025 · Registered: 08 March 2025 · Issued: 28 July 2025

Registrants: Charotar English Medium School, Shri Ketankumar P. Patel, Shri Kartikbhai Y. Patel, Shah Neel Atulkumar, Gandhi Keval Vipulkumar, Patel Yug Chiragkumar, Kariya Hitansh Kishorbhai, Maji Dev Laxman, Pathaan Shahnawazkhan Asifbhai, Christian Allen Stevenson

Research Paper: "Spring-Loaded Piezoelectric Generating Striker Device: Design, Methodology, and Applications" · Filed 27 Feb 2025